

Project: Data Lake Implementation using AWS Glue and Athena

Services: S3, Glue, Athena, IAM.

Description: Create a data lake on S3, catalog the data using AWS Glue, and perform queries using Athena. Configure IAM policies for secure access.

Skills: Data cataloging, serverless querying, data lake architecture.

Definition: Data Lake Implementation using AWS Glue and Athena is a project that involves building a serverless data lake on Amazon S3, cataloging the datasets using AWS Glue, and performing SQL-based queries on the data with Amazon Athena. Access and permissions are controlled through IAM policies to ensure secure data management.

Scope of the Project:

- Store structured and unstructured data in Amazon S3 as a scalable data lake.
- Catalog datasets automatically using AWS Glue.
- Query data serverless using Amazon Athena.
- Secure resources with IAM policies.
- Provide hands-on experience with serverless data lake architecture and analytics.
- Easily extendable for larger datasets or different file formats.

Methodology:

1. Manual Data Lake Method

- Uses S3, Glue, Athena, IAM.
- Provides hands-on understanding of serverless data lake creation, cataloging, and querying.
- Enables step-by-step configuration and testing of data upload, schema cataloging, and SQL-based queries using Athena.
- Utilizes Glue crawlers to automatically detect schema and create tables in the data catalog.
- Allows quick deployment within AWS Free Tier for demos or academic projects with small datasets.
- Cost-effective and ideal for beginner-friendly, proof-of-concept, or learning-focused projects.

2. Glue ETL-Based Data Lake Method

- Uses S3, Glue (Crawler + ETL), Athena, IAM.
- Provides practical experience with automated data transformation and serverless data lake pipelines.
- Enables step-by-step configuration of raw data ingestion, ETL processing, cataloging, and querying using Athena.
- Utilizes Glue ETL jobs to transform and process data before storing in the processed folder.
- Reduces manual errors and ensures consistent data processing.
- Suitable for small-scale production-style, resume-worthy, or learning-focused projects.

AWS Services used in the Project:

1. Amazon S3

- **Definition:** Object storage service that allows storing and retrieving any amount of data.
- **Purpose:** Provides scalable, durable, and secure storage for datasets.
- **Role in Project:** Stores raw and processed data as well as Athena query results. Acts as the central repository for the data lake.

2.AWS Glue

- **Definition:** Fully managed ETL (Extract, Transform, Load) and data catalog service.
- **Purpose:** Catalog metadata, discovers schema, and optionally transforms data.
- **Role in Project:** Creates tables in the Glue Data Catalog for datasets and runs ETL jobs to process raw data into processed formats for analysis.

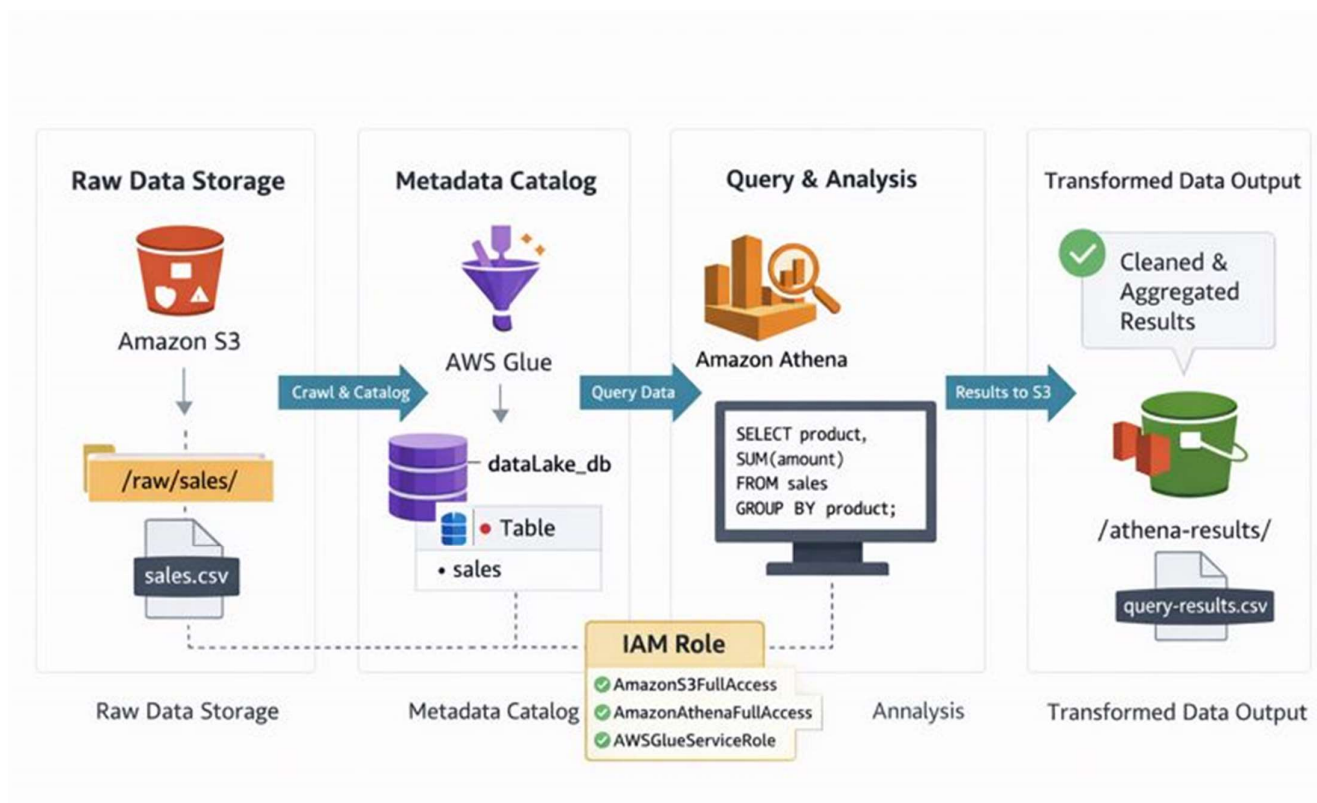
3.Amazon Athena

- **Definition:** Serverless SQL query service that allows analyzing data directly in S3 without managing servers.
- **Purpose:** Performs fast, serverless queries on large datasets.
- **Role in Project:** Executes SQL queries on S3 data and stores query results back in S3 for reporting or further analysis.

4.AWS IAM

- **Definition:** Identity and Access Management service to control access and permissions for AWS resources.
- **Purpose:** Ensures secure access and manages which users or services can interact with resources.
- **Role in Project:** Provides secure access control for S3, Glue, and Athena, ensuring only authorized users or services can use the data lake.

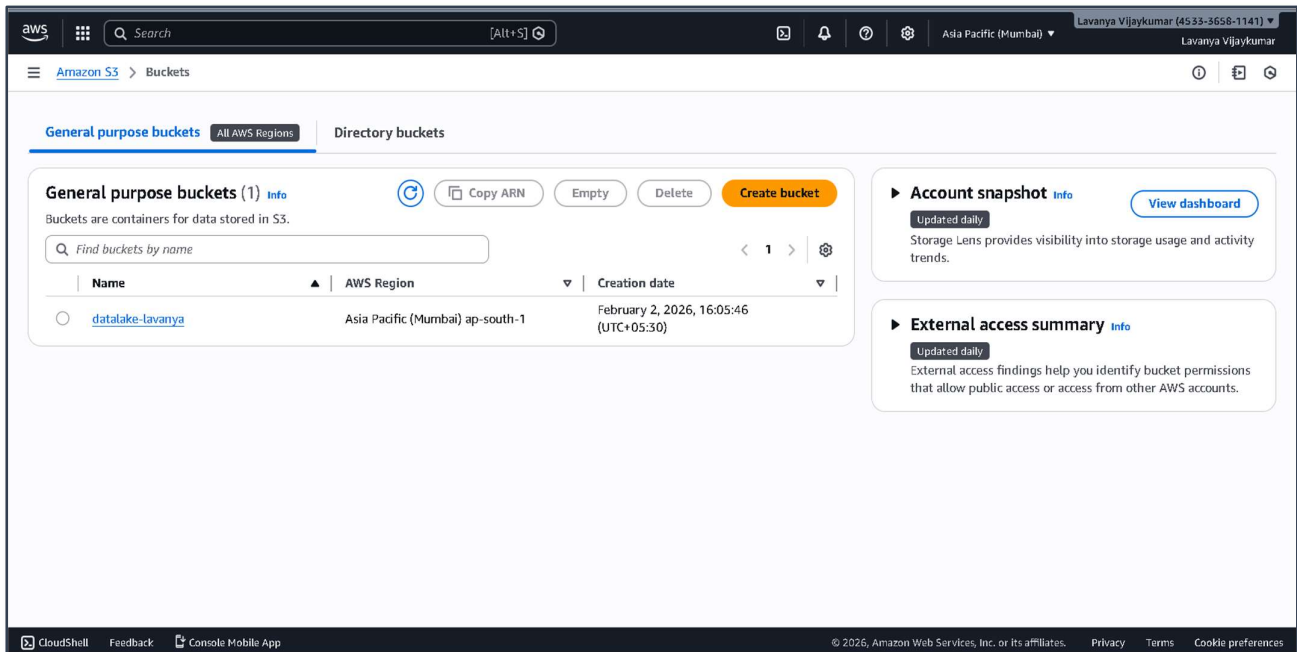
Architecture Diagram:



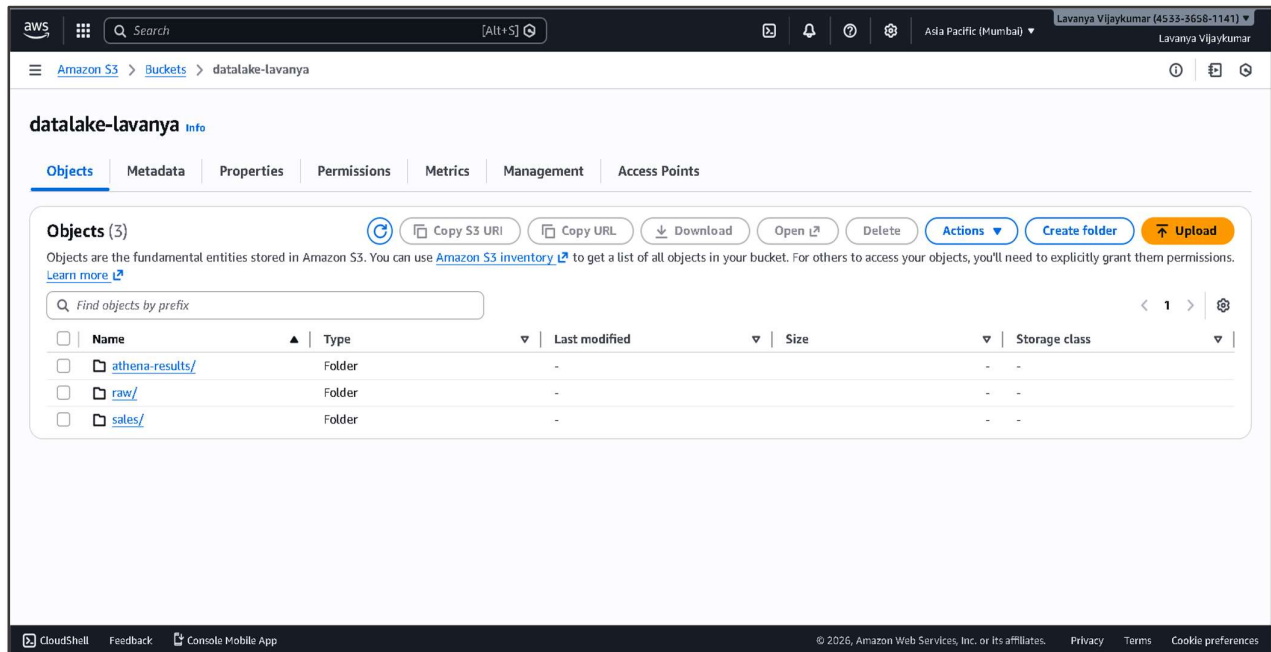
Project Implementation Steps:

STEP 1: Create S3 Bucket

- Open AWS S3 console.
- Click Create bucket.
- Name: datalake-lavanya
- Block public access
- Create Bucket

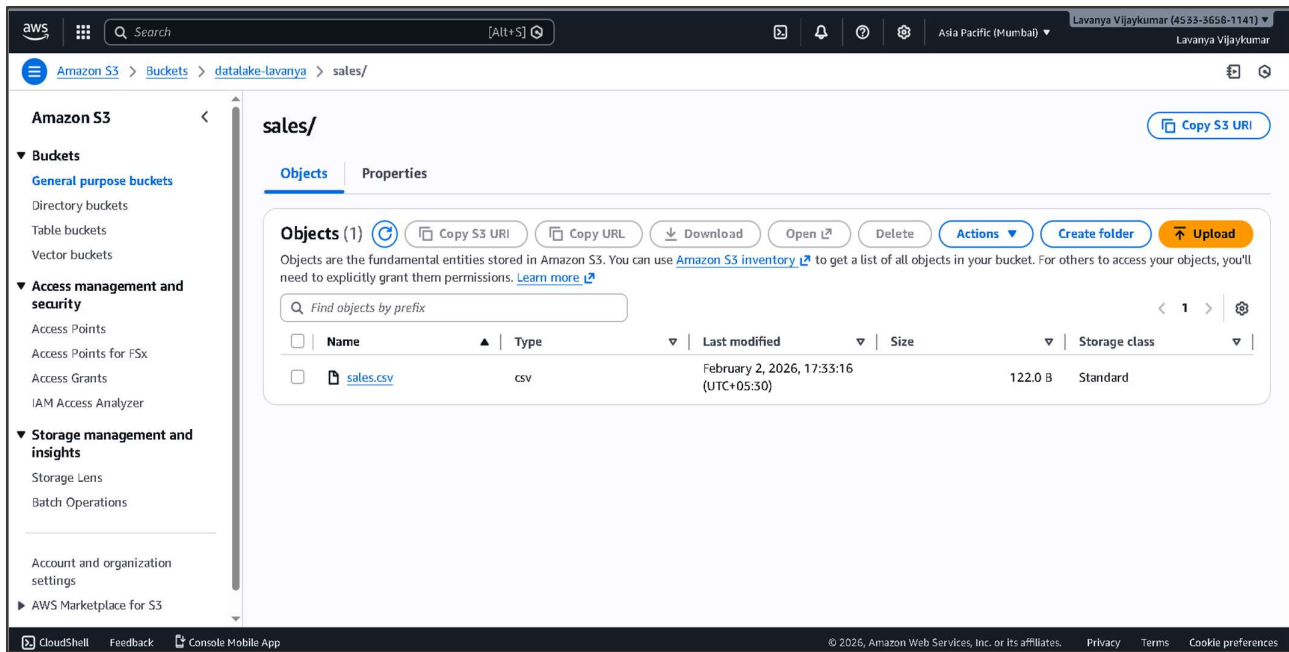


- Create folders inside the bucket:
 1. raw/
 2. athena-results/ → Store query results from Athena
 3. sales/ → Store original datasets



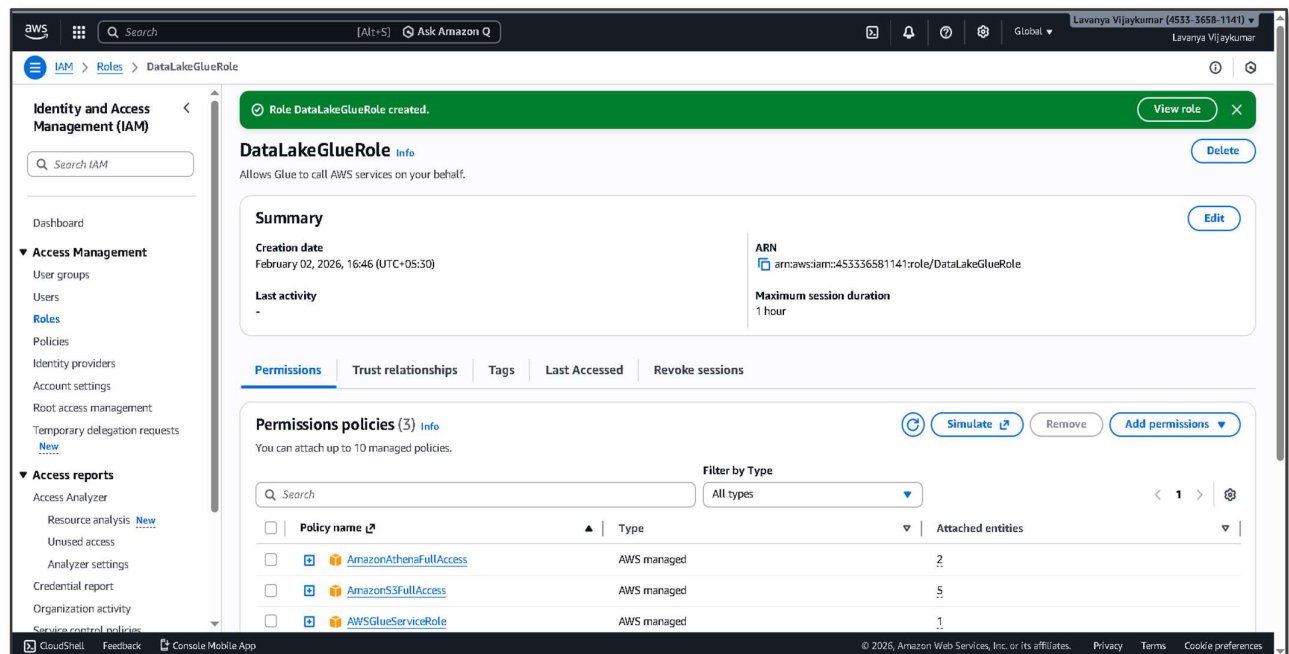
STEP 2: Upload Sample Data

- Go to the sales/ folder in S3.
- Upload datasets sales.csv file.



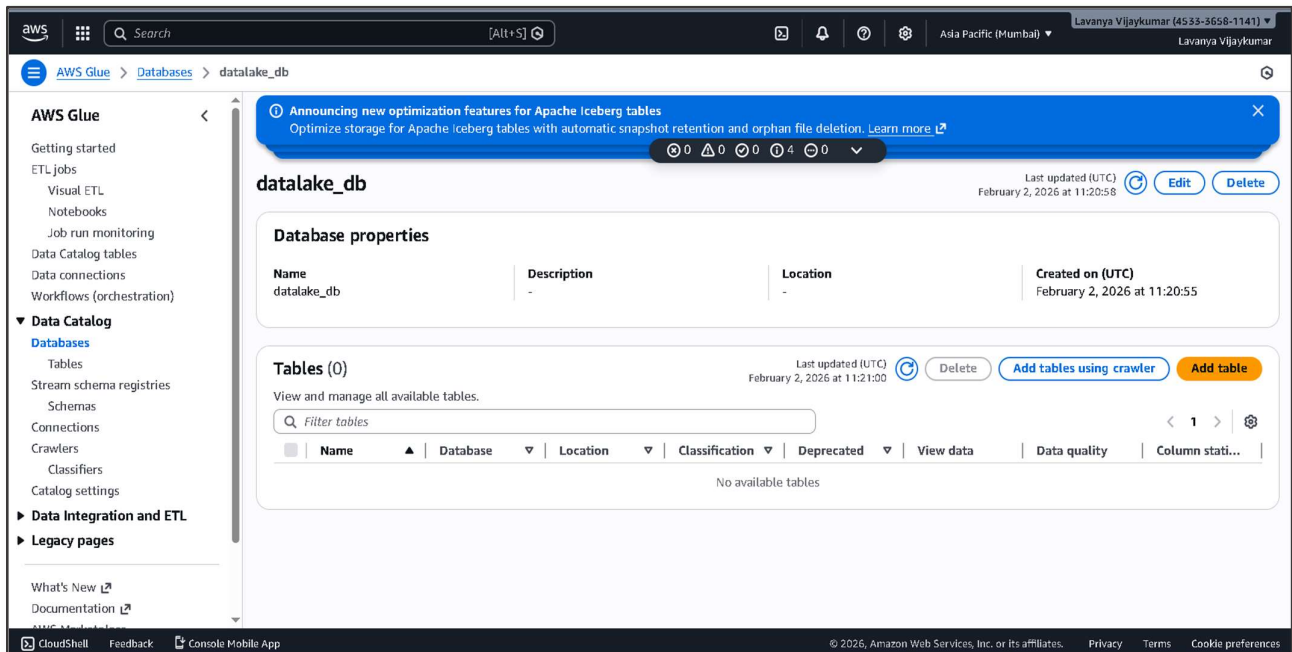
STEP 3: Set Up IAM Role for Glue and Athena

- Open IAM → Roles → Create Role
- Select AWS Service → Glue (for crawler and ETL)
- Attach policies:
 1. AWSGlueServiceRole → Required for Glue
 2. AmazonS3FullAccess → For accessing S3
 3. AmazonAthenaFullAccess → For Athena queries
- Name the role: DataLakeGlueRole
- Click Create role



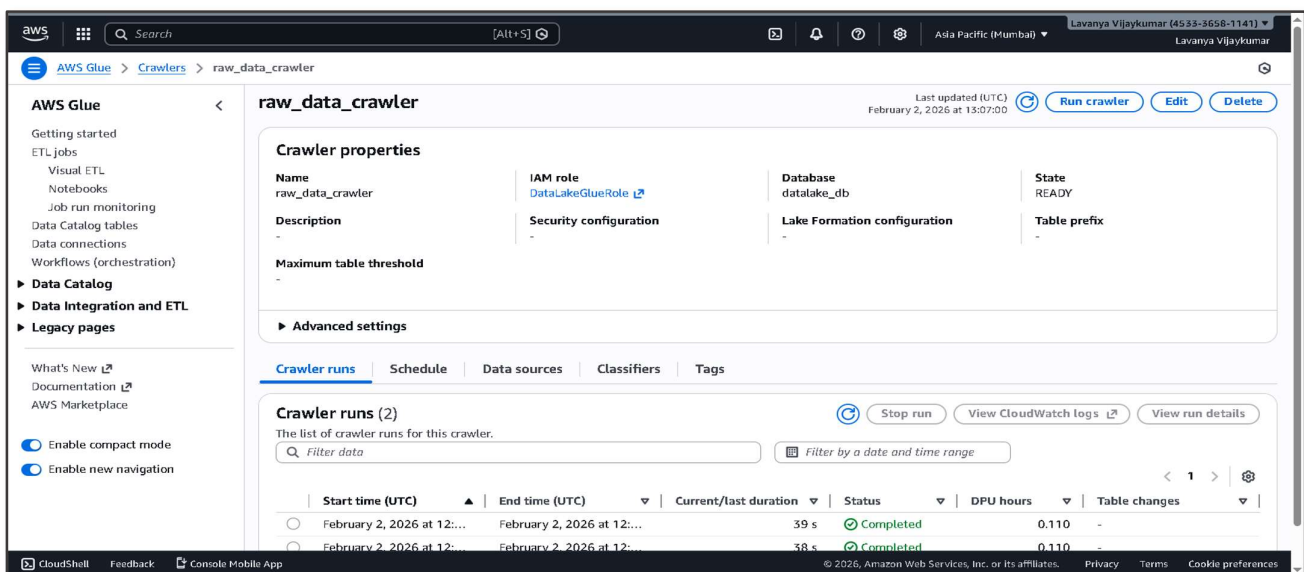
STEP 4: Create Glue Database

- Open AWS Glue → Databases → Add database
- Name it: datalake_db
- Save the database



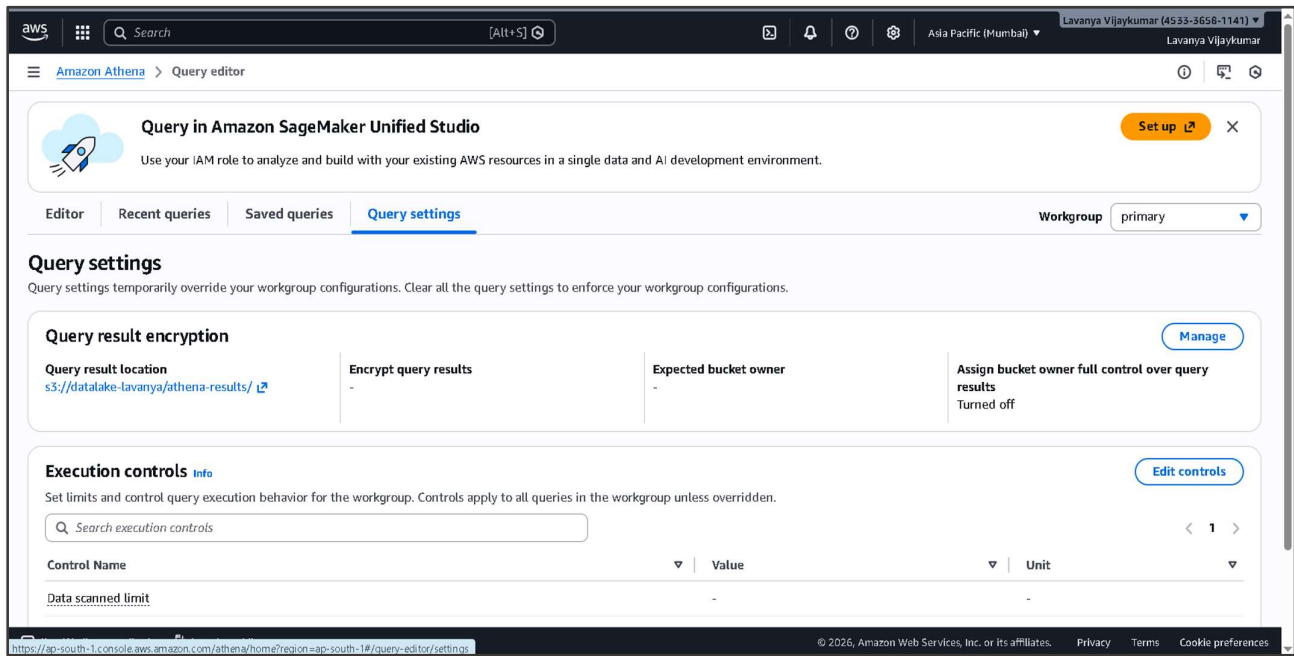
STEP 5: Create Glue Crawler

- Open AWS Glue → Crawlers → Add crawler
- Name: raw_data_crawler
- Data source: Select S3 → s3://datalake-lavanya/raw/sales/
- IAM Role: Select DataLakeGlueRole
- Set output to Glue database → datalake_db
- Add CSV classifier if your file has headers and comma separator
- Save crawler
- Run crawler and wait for Status = Ready



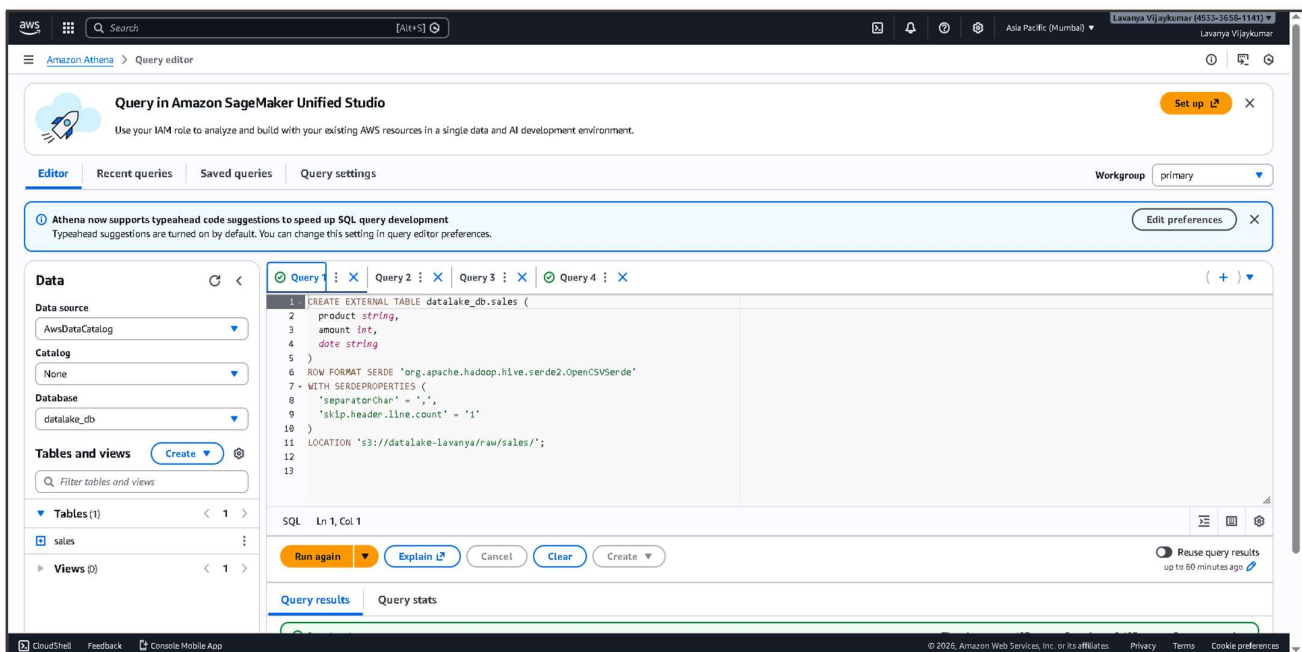
STEP 6: Set Athena Query Result Location

- Open Athena → Query Editor
- Set Query result location: `s3://datalake-lavanya/athena-results/`
- Click Save

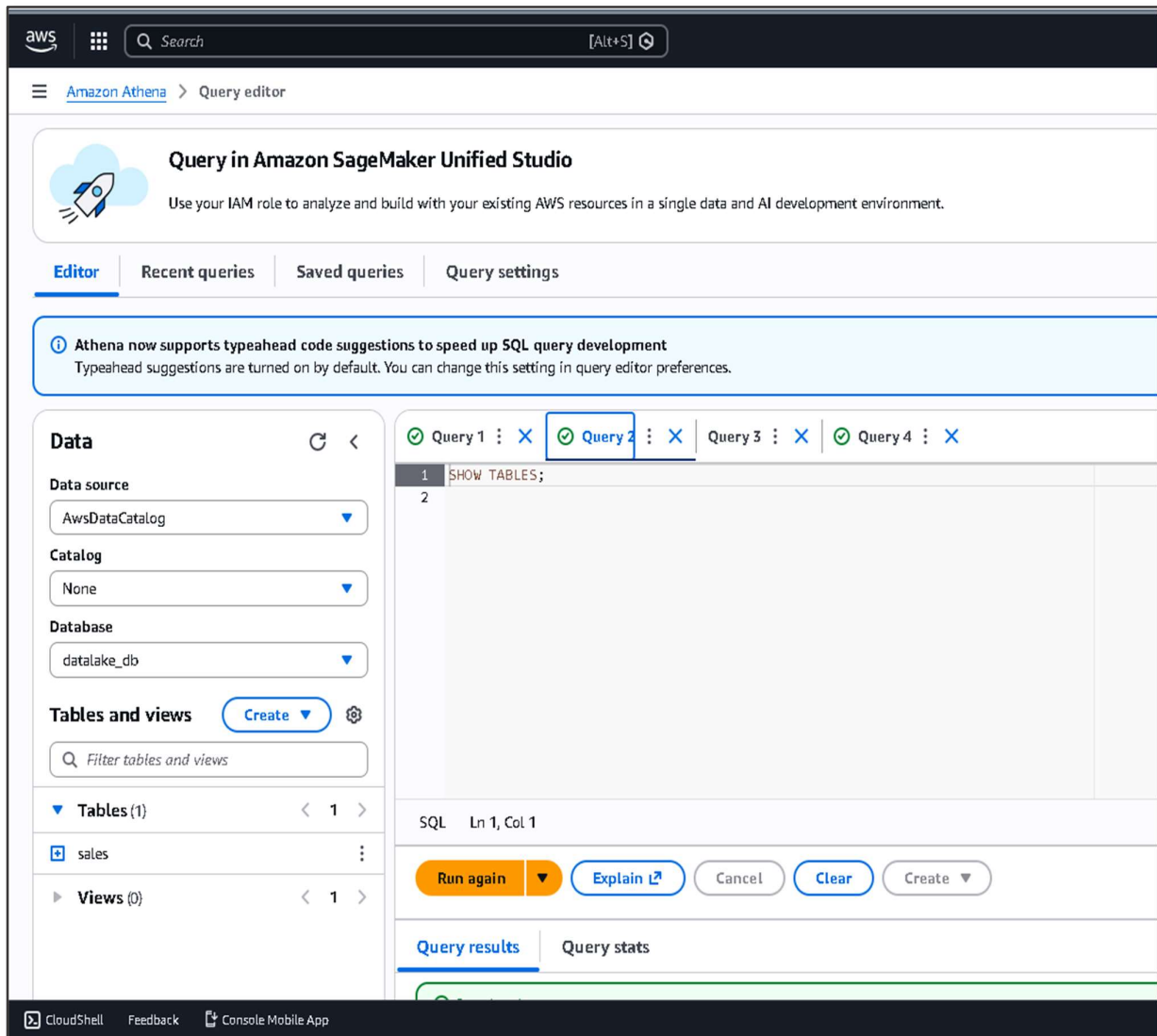


STEP 7: Create Table in Athena

- In Athena → Query Editor, run:

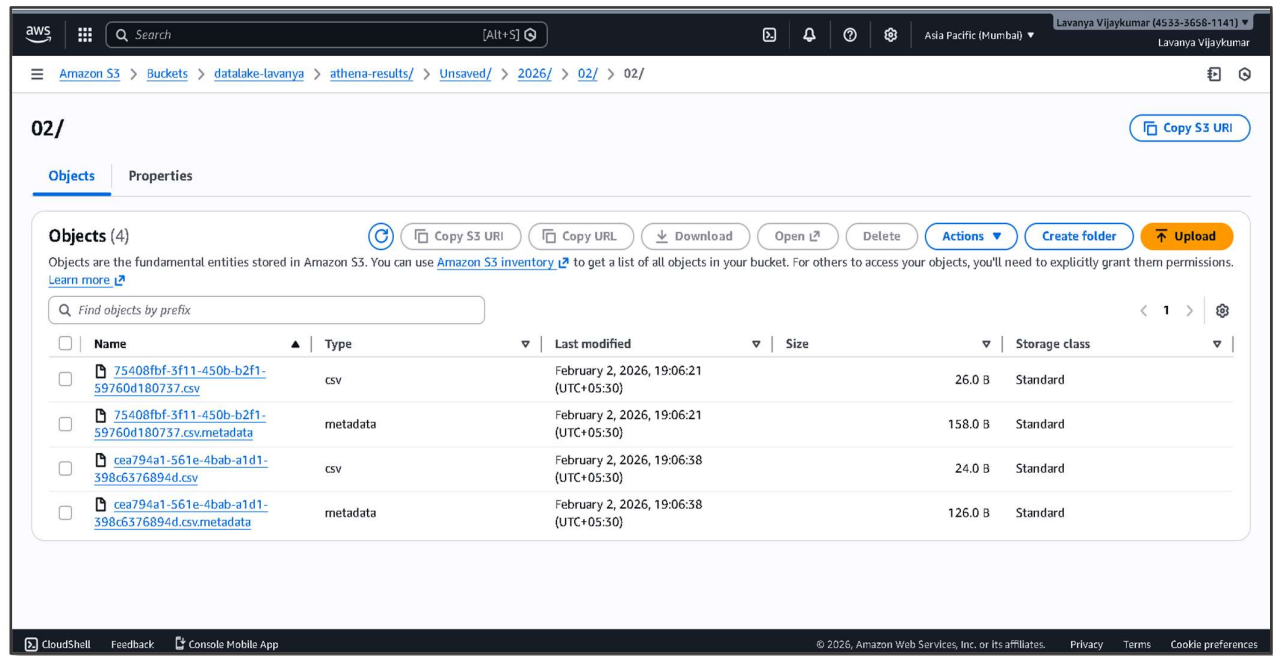


- Verify table creation:



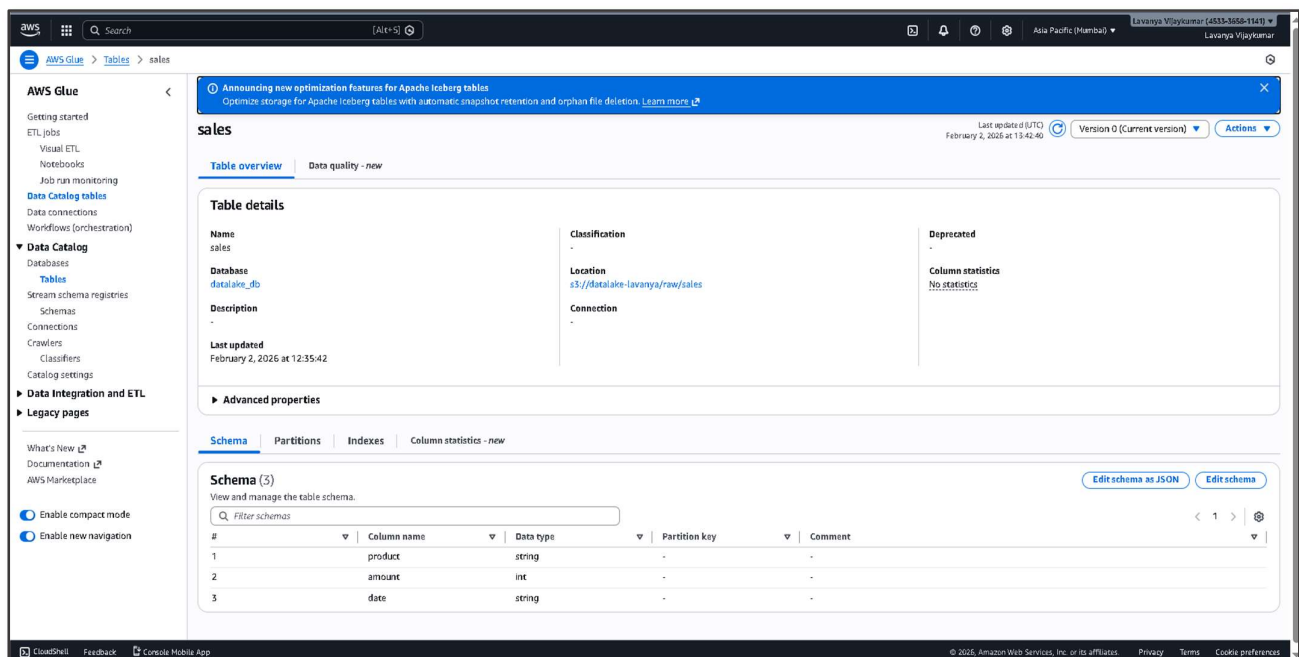
STEP 8: Query Data Using Athena

- Queries:
- `SELECT * FROM datalake_db. sales LIMIT 10;`
- `SELECT product, SUM (amount) AS total_sales
FROM datalake_db. sales
GROUP BY product;`
- Results stored in S3 (athena-results/ folder)



STEP 9: Verify Table Metadata in Glue

- Glue → Data Catalog → Tables → datalake_db → sales



Troubleshooting:

Issue: Table Not Found in Athena

Cause: Glue crawler did not create a table because of incorrect folder structure or missing classifier.

Solution:

- Ensure CSV is inside a subfolder in S3, e.g., s3://datalake-lavanya/raw/sales/sales.csv
- Add or use a proper CSV classifier in Glue crawler
- Re-run the crawler
- Verify table exists in Glue Data Catalog

Conclusion:

The Data Lake Implementation using AWS Glue and Athena project demonstrates a serverless, scalable data lake on AWS. Raw data is securely stored in S3, organized for easy management and future scalability.

AWS Glue catalog metadata, enabling schema discovery and easy querying, while Athena allows fast SQL queries on the raw data without altering it. Query results are stored as cleaned and transformed outputs in S3.

IAM roles ensure secure access across S3, Glue, and Athena, reinforcing cloud security best practices.

Overall, the project showcases a cost-effective, scalable, and manageable data lake, providing hands-on experience in data storage, cataloging, querying, and secure cloud architecture.